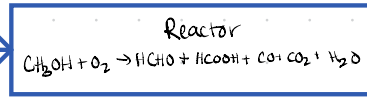


P3.24 Formaldehyde (HCHO) is produced by partial oxidation of methanol (CH₃OH). Several side reactions also occur, producing formic acid (HCOOH), CO, CO₂, and H₂O. Suppose 100 kgmol/h methanol and 20 kgmol/h O₂ are fed to a reactor, where 40% of the methanol and 95% of the O₂ is converted to products. What must the reactor outlet composition be?

M = Methanol
O = Oxygen gas
F = Formaldehyde
A = Formic Acid
C = Carbon Monoxide
D = Carbon Dioxide
W = Water

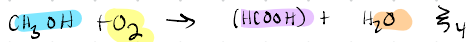
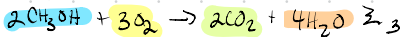
$$\begin{aligned}\dot{n}_M &= 100 \text{ kgmol/h} \\ \dot{n}_O &= 20 \text{ kgmol/h} \\ \dot{n}_{in} &= 120 \text{ kgmol/hr}\end{aligned}$$



$$\begin{aligned}\dot{n}_{M,o} &= 60 \text{ kg mol/hr} \\ \dot{n}_{O,o} &= 1 \text{ kg mol/hr} \\ \dot{n}_{F,o} &= 40 \text{ kgmol/hr} \\ \dot{n}_{A,o} &= 0 \\ \dot{n}_{C,o} &= 0 \\ \dot{n}_{D,o} &= 0 \\ \dot{n}_{W,o} &= 40 \text{ kg mol/hr}\end{aligned}$$



Reactions



Mole Balances & Recovery Equations

$$\dot{n}_{M,o} = 0.6 \dot{n}_{M,in} = 60 \text{ kgmol/h}$$

$$\dot{n}_{O,o} = 0.05 \dot{n}_{O,i} = 1 \text{ kgmol/h}$$

$$40 = 2\xi_1 + \xi_2 + 2\xi_3 + \xi_4$$

$$-40 = -2\xi_1 - 2\xi_2 - 6\xi_3 - 2\xi_4$$

$$0 = -\xi_2 - 4\xi_3 - \xi_4$$

This cannot equal to zero w/ ξ_2, ξ_3, ξ_4 all need to be positive

The only extent of reaction that is of importance is ξ_1 .

Reaction Equations

Methanol: $\dot{n}_{M,out} = \dot{n}_{M,in} - 2\xi_1 - \xi_2 - 2\xi_3 - \xi_4$

$$60 = 100 - \xi_1 - \xi_2 - \xi_3 - \xi_4$$

$$40 = 2\xi_1 + \xi_2 + 2\xi_3 + \xi_4$$

Oxygen: $\dot{n}_{O,out} = \dot{n}_{O,in} - \xi_1 - \xi_2 - 3\xi_3 - \xi_4$

$$1 = 19 - \xi_1 - \xi_2 - 3\xi_3 - \xi_4$$

$$\Rightarrow 2(20 = \xi_1 + \xi_2 + 3\xi_3 + \xi_4)$$

25 mg

Formaldehyde: $\dot{n}_{F,out} = \dot{n}_{F,in} + 2\xi_1$

Methanol: $\dot{n}_{M,out} = \dot{n}_{M,in} - 2\xi_1$

$$60 \text{ kgmol/h} = 100 \text{ kgmol/h} - 2\xi_1$$

$$2\xi_1 = 40 \text{ kgmol/h}$$

$$\xi_1 = 20 \text{ kgmol/h}$$

Water: $\dot{n}_{W,out} = \dot{n}_{W,in} + 2\xi_1$

$$\dot{n}_{W,out} = 2\xi_1 = 40 \text{ kgmol/h}$$

Total moles out: $\dot{n}_{M,out} + \dot{n}_{O,out} + \dot{n}_{F,out} + \dot{n}_{W,out}$
 $60 \text{ kgmol/h} + 1 \text{ kgmol/h} + 40 \text{ kgmol/h} + 40 \text{ kgmol/h}$

$$\dot{n}_{out, \text{total}} = 141 \text{ kgmol/h}$$

$$X_M = \frac{60}{141} = 0.426$$

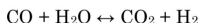
$$X_O = \frac{1}{141} = 0.007$$

$$X_F = \frac{40}{141} = 0.284$$

$$X_W = \frac{40}{141} = 0.284$$



(2) At low to moderate pressures, the equilibrium state of the water-gas shift reaction



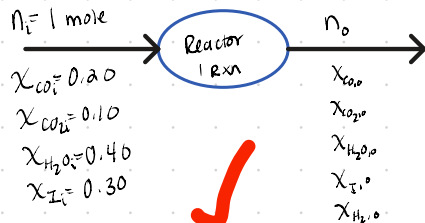
Is approximately described by the relation

$$\frac{x_{\text{CO}_2} x_{\text{H}_2}}{x_{\text{CO}} x_{\text{H}_2\text{O}}} = K_e(T) = a \exp\{b/T\}; \quad a = 0.0247 \text{ and } b = 4020 \text{ K}$$

where T is the reactor temperature in Kelvin, K_e is the reaction equilibrium constant, and x_i are the mole fractions of species i in the reactor at equilibrium. The feed to the reactor contains 20 mole % CO, 10 mole % CO₂, 40 mole % water, and the balance is inert gas. The reactor is maintained at $T = 1123 \text{ K}$.

- Assume a basis of 1 mole of feed and draw and label a flowchart. Carry out a degree-of-freedom analysis of the reactor based on the extent of reaction.
- Calculate the total moles of gas in the reactor at equilibrium and then the equilibrium mole fraction of hydrogen in the product.
- Suppose a gas sample is drawn from the reactor and analyzed shortly after startup and the mole fraction of hydrogen is significantly different from the calculated value. Assuming the calculations are correct, what is a likely explanation for the discrepancy?
- Use a method numerical method of your choice to take the input of the reactor temperature and the input feed component mole fractions of CO, H₂O, and CO₂ (assume no hydrogen is fed) and to calculate the mole fraction x_{H_2} in the product gas when equilibrium is reached.

Present your results in a table. Use your numerical method to try and maximize the yield of hydrogen by adjusting the temperature and feed composition.



a) DoF Analysis

$$\begin{array}{r} 4a - 9 \\ 4b - 1 \\ \hline 10 \end{array}$$

$$\begin{array}{r} 5a - 1 \\ 5b - 3 \quad (4-1) \\ 5c - 1 \quad \text{equilibrium} \\ \hline 5d = 5 \\ 10 \end{array}$$

10 - 10 = 0 well specified

b) $T = 1123 \text{ K}$ $K_e = a e^{b/T} = 0.0247 e^{4020/1123} = 0.8858$

$$K_e = \frac{x_{\text{CO}_2} x_{\text{H}_2}}{x_{\text{CO}} x_{\text{H}_2\text{O}}} = \frac{(0.1 + x_{\text{H}_2,e}) x_{\text{H}_2,e}}{(0.2 - x_{\text{H}_2,e})(0.4 - x_{\text{H}_2,e})}$$

$$K_e(0.08 - 0.4x_{\text{H}_2,e} + x_{\text{H}_2,e}^2) = 0.1x_{\text{H}_2,e} + x_{\text{H}_2,e}^2$$

$$0.1142x_{\text{H}_2,e}^2 + 0.63148x_{\text{H}_2,e} - 0.070864 = 0$$

$$-0.63148 \pm 0.6566$$

$$0.2284$$

$$x_{\text{H}_2,e} = 0.1100$$

At equilibrium:

$$x_{\text{CO}_2} = x_{\text{CO}_2,i} + x_{\text{H}_2,e} = 0.1 + x_{\text{H}_2,e}$$

$$x_{\text{H}_2\text{O}} = x_{\text{H}_2\text{O},i} - x_{\text{H}_2,e} = 0.4 - x_{\text{H}_2,e}$$

$$x_{\text{CO}} = x_{\text{CO},i} - x_{\text{H}_2,e} = 0.2 - x_{\text{H}_2,e}$$

$$x_{\text{CO}_2,e} = 0.1 + 0.1100 = 0.2100$$

$$x_{\text{H}_2\text{O},e} = 0.4 - 0.1100 = 0.2900$$

$$x_{\text{CO},e} = 0.2 - 0.1100 = 0.0900$$

- c) Right after start up, the process has not yet reached steady state. Equilibrium is not reached until steady state has been achieved. Therefore any sample taken during a period other than steady state will produce results inconsistent with equilibrium.

b= 4020		a= 0.0247	xco= 0.2	xH2O= 0.4	xCO2= 0.1
Temperature	Ke	xH2 at eq	(1-Ke)x^2+(0.1+.6Ke)x-0.08Ke=0		
100	7.10126E+15	#NUM!			
200	13243904.47	#NUM!			
300	16302.07965	#NUM!			
400	571.9479351	#NUM!			
500	76.6345458	#NUM!			
600	20.06642388	#NUM!			
700	7.705274601	#NUM!			
800	3.758605326	#NUM!			
900	2.150530662	#NUM!			
1000	1.375817314	#NUM!			
990	1.432833605	#NUM!			
1090	0.987203934	0.113835			
1190	0.724118103	0.102919			
1290	0.557288084	0.093691			
1390	0.445362953	0.085884			
1490	0.366790126	0.079251			

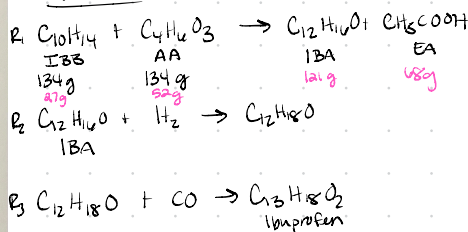
Goal seek



25/25

P4.10 Your company is interested in building a plant to make ibuprofen, using the new catalytic reaction scheme described in Example 4.1. Your job is to measure the reaction rate with different catalysts, as a first step in designing a full-scale reactor. In one experiment, you mix 134 g isobutylbenzene (IBB, $C_{10}H_{14}$) with 134 g acetic anhydride (AA, $C_4H_6O_3$) in a laboratory-scale batch reactor, adjust the temperature, add some catalyst, and wait 1 h. At the end of the hour you stop the reaction, collect all the material in the pot, and send it for chemical analysis. The chemist reports back that the pot contains 27 g IBB, 52 g acetic anhydride, 121 g isobutylacetophenone (IBA, $C_{12}H_{16}O$), and 68 g acetic acid (AAc, CH_3COOH). When you see these results you tell the chemist that the analysis is wrong. He protests. Who's right?

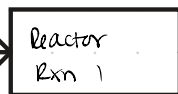
Reactions



Molar Masses

$$\begin{aligned}
 \text{IBB} &= 120 + 14 = 134 \text{ g/mol} \\
 \text{AA} &= 48 + 16 + 48 = 102 \text{ g/mol} \\
 \text{IBA} &= 144 + 16 + 16 = 176 \text{ g/mol} \\
 \text{EA} &= 12 + 4 + 32 = 48 \text{ g/mol}
 \end{aligned}$$

$$\begin{aligned}
 m_{\text{IBB}} &= 134 \text{ g} \\
 n_{\text{IBB}} &= 1 \text{ mole} \\
 m_{\text{AA}} &= 134 \text{ g} \\
 n_{\text{AA}} &= 1.314 \text{ mol} \\
 m_{\text{IBB, in}} &= 268 \text{ g}
 \end{aligned}$$



using 27 g of IBB to test the rest of the composition

$$n_{\text{IBB, out}} = 0.2015$$

$$\begin{aligned}
 m_{\text{IBB, out}} &= 27 \text{ g} \\
 m_{\text{IBB}} &= 27 \text{ g} \\
 m_{\text{AA}} &= 52 \text{ g} \\
 m_{\text{IBA}} &= 121 \text{ g} \\
 m_{\text{EA}} &= 68 \text{ g}
 \end{aligned}$$

$$n_{\text{IBB, out}} = n_{\text{IBB, in}} - \xi_1$$

$$\begin{aligned}
 \xi_1 &= n_{\text{IBB, in}} - n_{\text{IBB, out}} \\
 &= 1.000 \text{ mol} - 0.2015 \text{ mol} \\
 &= 0.7985 \text{ mol}
 \end{aligned}$$

$$\begin{aligned}
 n_{\text{AA, out}} &= n_{\text{AA, in}} - \xi_1 \\
 &= 1.314 - 0.7985 \\
 &= 0.5155 \text{ mol}
 \end{aligned}$$

$$\begin{aligned}
 n_{\text{IBA, out}} &= n_{\text{IBA, in}} + \xi_1, \quad m_{\text{IBA}} = (n_{\text{IBA}})(M_{\text{IBA}}) \\
 &= 0.7985 \text{ mol} \quad = 140.53 \text{ g}
 \end{aligned}$$

$$\begin{aligned}
 n_{\text{EA, out}} &= n_{\text{EA, in}} + \xi_1, \quad m_{\text{EA}} = (n_{\text{EA}})(M_{\text{EA}}) \\
 &= 0.7985 \text{ mol} \quad = 38.328 \text{ g}
 \end{aligned}$$

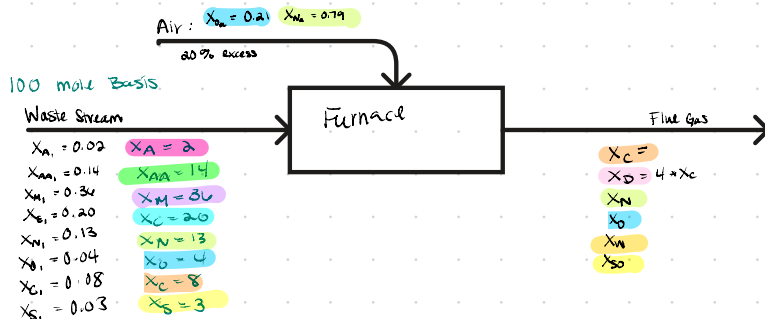
Assuming the output of any one of the output streams is true, the calculations show that the chemist's analysis is not possible based on the extent of reaction. The calculated mass differs greatly from the chemist's analysis.



25/25

P4.19 Waste streams from various processes in a complex chemical facility are combined and burned for heat, because the cost of purifying and recycling the components is prohibitive. On a typical day, the combined waste stream contains 2 mol% acetone $(\text{CH}_3)_2\text{CO}$, 14 mol% acetic anhydride $(\text{CH}_3\text{CO})_2\text{O}$, 36 mol% methane CH_4 , 20 mol% ethane C_2H_6 , 13 mol% N_2 , 4 mol% O_2 , 8 mol% CO , and 3 mol% H_2S . The waste stream is fed to a furnace along with 20% excess air and combusted. (For combustion reactions, excess air is calculated on the assumption that the only products of combustion are CO_2 , H_2O , N_2 , and SO_2 .) Analysis of the exit flue gas gives a CO_2/CO mass ratio of 4:1. The only other components in the flue gas are N_2 , O_2 , H_2O , and SO_2 . Calculate the composition of the exit flue gas. Assume air is 21 mol% O_2 and 79 mol% N_2 .

A = Acetone $(\text{CH}_3)_2\text{CO}$
 AA = Acetic Anhydride $(\text{CH}_3\text{CO})_2\text{O}$
 M = Methane CH_4
 E = Ethane C_2H_6
 N = Nitrogen N_2
 C = Carbon Monoxide CO
 S = dihydrogen Sulfide H_2S
 O = Oxygen gas O_2
 D = Carbon Dioxide CO_2
 W = Water H_2O
 SO = Sulfur dioxide SO_2



Atomic Balances

Carbon: $3X_A + 4X_{AA} + X_M + 2X_E + X_C = X_{\text{C}_2} + X_{\text{D}} \quad X_{\text{D}} = 4X_C$

Oxygen: $X_A + 3X_{AA} + 2X_O + 2X_{\text{O}_2} + X_C = X_C + 2X_{\text{D}_2} + 2X_{\text{O}_3} + X_W + 2X_{\text{SO}}$

Hydrogen: $6X_A + 6X_{AA} + 4X_M + 6X_E + 2X_S = 2X_W$

Nitrogen: $2X_N + 2X_{\text{N}_2} = 2X_N$

Sulfur: $X_S = X_{\text{SO}}$

① Solve for Hydrogen first since there are no hydrogens in the air stream

$$6(2) + 6(14) + 4(36) + 6(20) + 2(8) = 2X_W$$

$$376 = 2X_W$$

$$188 = X_W$$

② Solve for Sulfur

$$3 = X_{\text{SO}}$$

③ Solve for carbon

$$3(2) + 4(14) + 36 + 2(20) + 8 = 5X_{\text{C}_2}$$

$$146 = 5X_{\text{C}_2}$$

$$29.2 = X_{\text{C}_2}$$

$$X_{\text{D}_2} = 116.8$$

④ Solve for oxygen without the excess air

$$2 + 3(14) + 4(2) + 2X_{O_2} + 8 = 29.2 + 2(116.8) + 2X_{O_3} + 188 + 2(3)$$

$$60 + 2X_{O_2} = 456.8 + 2X_{O_3}$$

$$\text{needed } O_2 \text{ from air} = \frac{456.8 - 60}{2} = 198.4$$

excess O_2 is 1.2 times what is needed:

$$X_{O_2} = (1.2)(198.4) = 238.08$$

$$60 + 2(238.08) = 456.8 + 2X_{O_3}$$

$$X_{O_3} = 39.68$$

⑤ Solve for nitrogen

$$X_{N_2} = 0.79 \left(\frac{238.08}{0.21} \right) = 895.4$$

$$X_{N_3} = X_{N_1} + X_{N_2} = 13 + 895.6 = 908.6$$

Total moles out based on theoretical 100 mole basis of waste stream:

$$X_{C_2} + X_D + X_{N_3} + X_{O_3} + X_W + X_{S_0}$$

$$29.2 + 116.8 + 908.4 + 39.68 + 188 + 3 = 1285.3$$

mol composition of outlet stream

Carbon monoxide: $X_{C_2} = 0.0227$

Carbon dioxide: $X_D = 0.0909$

nitrogen: $X_{N_3} = 0.7069$

oxygen: $X_{O_3} = 0.0309$

Water: $X_W = 0.1463$

Sulfur dioxide: $X_{S_0} = 0.0023$

✓ 25/25